

Displays

D1:

$$(1 - \varepsilon)\mu(F(\mathcal{X})) \leq \mu(F(\mathcal{S})) \leq \mu(F(\mathcal{X})). \quad (1)$$

D2:

$$d_H(A, B) = \max \left\{ \sup_{a \in A} \inf_{b \in B} \|a - b\|, \sup_{b \in B} \inf_{a \in A} \|a - b\| \right\},$$

D3:

$$d_H(A, B) = \inf \{ \delta \geq 0 : A \subseteq B_\delta \text{ and } B \subseteq A_\delta \},$$

D4:

$$\left(\frac{1}{\delta}\right)^d \frac{\mu(\mathcal{X})}{C} \leq N(\mathcal{X}, \delta) \leq M(\mathcal{X}, \delta) \leq \left(\frac{2\Delta(\mathcal{X})}{\delta} + 1\right)^d,$$

D5:

$$\left(\frac{1}{\delta}\right)^d \frac{\mu(\mathcal{X})}{C} \leq |\mathcal{S}| \leq \left(\frac{3\Delta(\mathcal{X})}{\delta}\right)^d,$$

D6:

$$|\mathcal{S}| \geq N(\mathcal{X}, \delta) \geq \left(\frac{1}{\delta}\right)^d \frac{\mu(\mathcal{X})}{C}.$$

D7:

$$\lambda(\partial A) = \liminf_{\delta \rightarrow 0} \frac{\mu(A_\delta) - \mu(A)}{\delta}. \quad (2)$$

D8:

$$c = \max\{M/(d\mu(\mathcal{B}_1(\cdot))^{1/d}), 1\} < \infty,$$

D9:

$$\frac{\lambda(\partial A_\delta)}{\mu(A_\delta)^{(d-1)/d}} \leq \frac{c \lambda(\partial A)}{\mu(A)^{(d-1)/d}},$$

D10:

$$\mu(A_\delta) \leq \left(1 + \frac{c \delta \lambda(\partial A)}{\mu(A) d}\right)^d \mu(A), \quad (3)$$

D11:

$$\begin{aligned} \lambda(\partial A) &= \liminf_{\delta \rightarrow 0} \frac{\mu(A_\delta) - \mu(A)}{\delta} = \liminf_{\delta \rightarrow 0} h(\delta) \\ &= \limsup_{\delta \rightarrow 0} h(\delta) = \lim_{\delta \rightarrow 0} h(\delta). \end{aligned}$$

D12:

$$\mu(A_{\delta'}) < \mu(A) + \delta' (\lambda(\partial A) + \varepsilon) = \mu(A) + \delta' \lambda'.$$

D13:

$$\begin{aligned} g(\delta') &= \left(\frac{\mu(A_{\delta'})}{\mu(A)}\right)^{1/d} < \left(1 + \frac{\delta' \lambda'}{\mu(A)}\right)^{1/d} \leq 1 + \frac{\delta' \lambda'}{\mu(A) d} \\ &\leq 1 + \frac{c \delta' \lambda'}{\mu(A) d}, \end{aligned} \quad (4)$$

D14:

$$\begin{aligned} \frac{d g(\delta')}{d \delta'} &= \frac{1}{\mu(A)^{1/d}} \cdot \frac{d \mu(A_{\delta'})^{1/d}}{d \delta'} \\ &= \frac{\lambda(\partial A_{\delta'})}{\mu(A)^{1/d} \mu(A_{\delta'})^{(d-1)/d} d}, \end{aligned}$$

D15:

$$\begin{aligned} \frac{d}{d \delta'} \left(1 + \frac{c \delta' \lambda'}{\mu(A) d}\right) &= \frac{c \lambda'}{\mu(A) d} > \frac{c \lambda(\partial A)}{\mu(A) d} \\ &\geq \frac{1}{d \mu(A)^{1/d}} \cdot \frac{\lambda(\partial A_{\delta'})}{\mu(A_{\delta'})^{(d-1)/d}} = \frac{d g(\delta')}{d \delta'}, \end{aligned}$$

D16:

$$\begin{aligned} g(\delta) &= \lim_{\lambda' \rightarrow \lambda(\partial A)} g(\delta) \leq \lim_{\lambda' \rightarrow \lambda(\partial A)} \left(1 + \frac{c \lambda(\partial A) \delta}{\mu(A) d} \right) \\ &= 1 + \frac{c \delta \lambda(\partial A)}{\mu(A) d}, \end{aligned}$$

D17:

$$f(x) \subseteq F(\mathcal{B}_\delta(x)) \subseteq f(x)_{\delta K}, \quad (5)$$

D18:

$$d_H(f(x), f(y)) \leq K \|x - y\| \leq K \delta.$$

D19:

$$d_H(f(x), \cup_{y \in \mathcal{B}_\delta(x)} f(y)) = d_H(f(x), F(\mathcal{B}_\delta(x))) \leq \delta K.$$

D20:

$$F(\mathcal{X}) = \cup_{x \in \mathcal{S}} F(\mathcal{B}_\delta(x)) \subseteq \cup_{x \in \mathcal{S}} f(x)_{\delta K} = F(\mathcal{S})_{\delta K}, \quad (6)$$

D21:

$$F(\mathcal{X}) = F(\cup_{x \in \mathcal{S}} \cup_{y \in \mathcal{B}_\delta(x)} y) = \cup_{x \in \mathcal{S}} F(\mathcal{B}_\delta(x)).$$

D22:

$$\delta \leq \frac{d((1 - \varepsilon)^{-1/d} - 1)}{\alpha K c},$$

D23:

$$\frac{d}{dt} \begin{pmatrix} x \\ y \\ \theta \end{pmatrix} = \begin{pmatrix} u_v \cos \theta \\ u_v \sin \theta \\ u_\omega \end{pmatrix}, \quad (7)$$

D24:

$$\begin{pmatrix} x_{RSL} \\ y_{RSL} \\ \theta_{RSL} \end{pmatrix} = \rho \begin{pmatrix} 2 \sin(\theta_1) + \theta_2 \cos(\theta_1) - \sin(\theta_1 - \theta_3) \\ -1 + 2 \cos(\theta_1) - \theta_2 \sin(\theta_1) - \cos(\theta_1 - \theta_3) \\ \theta_1 / \rho - \theta_2 / \rho \end{pmatrix},$$

Inline

I1: *

I2: $(1 - \varepsilon) = 0.2$

I3: $(1 - \varepsilon) = 0.4$

I4: $(1 - \varepsilon) = 0.6$

I5: $(1 - \varepsilon) = 0.8$

I6: 0.2

I7: 0.4

I8: 0.6

I9: 0.8

I10: ε

I11: *

I12: $\dot{x} = h(x, u)$

I13: $x \in \mathbb{R}^d$

I14: $u \in \mathcal{U}$

I15: $\mathcal{U} \subset \mathbb{R}^m$

I16: h

I17: $\mathbf{x}(x_0, t, u(\cdot))$

I18: t

I19: x_0

I20: $u(t)$

I21: $H(x_0, T) = \{\mathbf{x}(x_0, T, u(\cdot)) \mid u(t) \in \mathcal{U}, \forall t \in [0, T]\}$

I22: $\mathcal{X} \subset \mathbb{R}^d$

I23: d

I24: $\mathcal{Y}$

I25: $f : \mathbb{R}^d \rightarrow 2^{\mathcal{Y}}$
 I26: $x \in \mathcal{X}$
 I27: $f(x) \subseteq 2^{\mathcal{Y}}$
 I28: $T > 0$
 I29: $f(x) = H(x, T)$
 I30: f
 I31: $f(z) = \emptyset \forall z \notin \mathcal{X}$
 I32: $\mathcal{X}' \subseteq \mathcal{X}$
 I33: $\mathcal{X}'$
 I34: $F(\mathcal{X}') = \cup_{x \in \mathcal{X}'} f(x)$
 I35: F
 I36: $F(\mathcal{X}') \subseteq F(\mathcal{X})$
 I37: $F(\mathcal{X})$
 I38: $\mathcal{X}$
 I39: $\mathcal{S} \subset \mathcal{X}$
 I40: $|\mathcal{S}| = n \in \mathbb{N}_+$
 I41: $F(\mathcal{S}) \approx F(\mathcal{X})$
 I42: $F(\mathcal{S})$
 I43: $\mu(\cdot)$
 I44: $\mu(F(\mathcal{X}))$
 I45: $\varepsilon \in (0, 1)$
 I46: $f(x)$
 I47: $\mathcal{S}$
 I48: δ
 I49: $\delta > 0$
 I50: $\delta \in \mathbb{R}_+$
 I51: $\mathcal{S} \leftarrow$
 I52: $\exists x \in \mathcal{X} : \forall y \in \mathcal{S}, \|x - y\| \geq \delta$
 I53: $\mathcal{S} \leftarrow \mathcal{S} \cup \{x\}$
 I54: $\hat{F}_{\mathcal{S}}$
 I55: $\mu(\hat{F}_{\mathcal{S}}) \geq (1 - \varepsilon)\mu(F(\mathcal{X}))$
 I56: $\alpha \leftarrow$
 I57: $(\mathcal{X})$
 I58: $K \leftarrow$
 I59: $\triangleright$
 I60: $c \leftarrow$
 I61: $\delta \leftarrow d \left((1 - \varepsilon)^{-1/d} - 1 \right) / (\alpha K c)$
 I62: $(\mathcal{X}, \delta)$
 I63: $\hat{F}_{\mathcal{S}} \leftarrow \emptyset$
 I64: $x \in \mathcal{S}$
 I65: $\hat{f}(x) \leftarrow$
 I66: (x)
 I67: $\hat{F}_{\mathcal{S}} \leftarrow \hat{F}_{\mathcal{S}} \cup \hat{f}(x)$
 I68: i
 I69: $\varepsilon \leftarrow 1/2$
 I70: $\hat{F}_{\mathcal{S}} \leftarrow$
 I71: $(\mathcal{X}, \varepsilon)$
 I72: $\varepsilon \leftarrow \varepsilon/2$
 I73: x
 I74: $(1 - \varepsilon)$
 I75: A, B
 I76: $d_{\text{H}}(A, B)$
 I77: A
 I78: B
 I79: $\|\cdot\|$
 I80: $A_{\delta} \subseteq \mathbb{R}^d$

I81: $A_\delta = \cup_{a \in A} \mathcal{B}_\delta(a)$
 I82: $\mathcal{B}_\delta(a)$
 I83: a
 I84: $\forall x \in \mathcal{X} \mu(f(x)) > 0$
 I85: $K > 0$
 I86: $x, y \in \mathcal{X}$
 I87: $d_H(f(x), f(y)) \leq K \|x - y\|$
 I88: $A \subseteq \mathcal{Y}$
 I89: m
 I90: $g : \mathbb{R}^m \rightarrow \mathcal{Y}$
 I91: $\mathcal{X}' \subseteq \mathcal{X} \subseteq \mathbb{R}^d$
 I92: $F(\mathcal{X}')$
 I93: $\delta \geq 0$
 I94: $F(\mathcal{X}')_\delta$
 I95: $(d - 1)$
 I96: $(A, \|\cdot\|)$
 I97: $C = \{c_1, \dots, c_N\} \subset B$
 I98: $B \subset A$
 I99: $b \in B$
 I100: $c \in C$
 I101: $\|b - c\| \leq \delta$
 I102: $N(B, \delta)$
 I103: $D = \{d_1, \dots, d_M\} \subset B$
 I104: $\min_{i,j \in [M]: i \neq j} \|d_i - d_j\| > \delta$
 I105: $M(B, \delta)$
 I106: $C = \frac{\pi^{d/2}}{\Gamma(d/2+1)}$
 I107: $\Delta(\mathcal{X}) = \max_{x,y \in \mathcal{X}} \|x - y\|$
 I108: $\Gamma(\cdot)$
 I109: $\mathcal{X} \subseteq \mathbb{R}^d$
 I110: $\delta \leq \Delta(\mathcal{X})$
 I111: $\delta > \Delta(\mathcal{X})$
 I112: $\delta \in (0, \Delta(\mathcal{X})]$
 I113: $\Delta(\mathcal{X})$
 I114: C
 I115: $\exists x \in \mathcal{X} \forall y \in \mathcal{S} \|x - y\| > \delta$
 I116: $\forall x \in \mathcal{X} \exists y \in \mathcal{S} \|x - y\| \leq \delta$
 I117: $|\mathcal{S}| \geq N(\mathcal{X}, \delta)$
 I118: $M(\mathcal{X}, \delta)$
 I119: $\blacksquare$
 I120: $A \subseteq \mathbb{R}^d$
 I121: $\lambda(\partial A)$
 I122: $A \subset \mathbb{R}^d$
 I123: $\mu(A) > 0$
 I124: A_δ
 I125: $c \geq 1$
 I126: $M > 0$
 I127: c
 I128: $\mu(A)$
 I129: $\mu(A_\delta)$
 I130: $g : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$
 I131: $g(x) = (\mu(A_x)/\mu(A))^{1/d}$
 I132: $h(\delta) : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$
 I133: $h(\delta) = \frac{\mu(A_\delta) - \mu(A)}{\delta}$
 I134: $\varepsilon > 0$
 I135: $\lambda' = \lambda(\partial A) + \varepsilon > \lambda(\partial A)$
 I136: $\lim_{\delta \rightarrow 0} h(\delta) = \lambda(\partial A)$

I137: $\xi(\varepsilon) > 0$
 I138: $\delta' \in (0, \xi(\varepsilon))$
 I139: $|h(\delta') - \lambda(\partial A)| < \varepsilon$
 I140: $h(\delta') < \lambda(\partial A) + \varepsilon$
 I141: $h(\delta')$
 I142: $\delta \in (0, \xi(\varepsilon))$
 I143: $\delta \in [\xi(\varepsilon), \Delta(\mathcal{X})]$
 I144: $g(\delta')$
 I145: δ'
 I146: $A_{\delta'}$
 I147: $\delta' \in (0, \Delta(\mathcal{X})]$
 I148: $g(\delta) \leq 1 + \frac{c\delta\lambda'}{\mu(A)d}$
 I149: $g(\delta)$
 I150: $f(x)_{\delta K}$
 I151: (δK)
 I152: K
 I153: $\mathcal{B}_\delta(x)$
 I154: $y \in \mathcal{B}_\delta(x)$
 I155: $x \in \mathcal{B}_\delta(x)$
 I156: $f(x) \subseteq F(\mathcal{B}_\delta(x))$
 I157: $f(y)$
 I158: $f(y) \subseteq f(x)_{\delta K}$
 I159: $\mathcal{S} \subset \mathcal{X} \subseteq \mathbb{R}^d$
 I160: $F(\mathcal{S})_{\delta K}$
 I161: $|\mathcal{S}|$
 I162: $\forall x \notin \mathcal{X}, f(x) = \emptyset$
 I163: $F(\mathcal{B}_\delta(x)) \subseteq f(x)_{\delta K}$
 I164: $\cup_{x \in \mathcal{S}} F(\mathcal{B}_\delta(x)) = \cup_{x \in \mathcal{S}} f(x)_{\delta K}$
 I165: $\cup_{x \in \mathcal{S}} f(x)_{\delta K} = F(\mathcal{S})_{\delta K}$
 I166: $\alpha = \sup_{\mathcal{X}' \subseteq \mathcal{X}} \frac{\lambda(\partial F(\mathcal{X}'))}{\mu(F(\mathcal{X}'))} < \infty$
 I167: $c = \max\{M/(d\mu(\mathcal{B}_1(\cdot))^{1/d}), 1\}$
 I168: $(1 - \varepsilon)\mu(F(\mathcal{X})) \leq \mu(F(\mathcal{S})) \leq \mu(F(\mathcal{X}))$
 I169: $\mu(\hat{F}_{\mathcal{S}}) \geq (1 - \varepsilon)\mu(F(\mathcal{S}))$
 I170: $|\mathcal{S}| \leq (3\alpha K \Delta(\mathcal{X})c/\varepsilon)^d$
 I171: $\mathcal{C}_\alpha$
 I172: $\mathcal{C}_K$
 I173: $\mathcal{C}_c$
 I174: α
 I175: $\mathcal{C}_{\mathcal{S}}$
 I176: $\mathcal{C}_f$
 I177: $\mathcal{O}(\mathcal{C}_\alpha + \mathcal{C}_K + \mathcal{C}_c + \mathcal{C}_{\mathcal{S}} + (\alpha K \Delta(\mathcal{X})c/\varepsilon)^d \mathcal{C}_f)$
 I178: $(\mathcal{F}_i)_{i \in \mathbb{N}_+}$
 I179: $i \in \mathbb{N}_+$
 I180: $\mathcal{F}_i$
 I181: $\hat{F}_{\mathcal{S}_i}$
 I182: $\mathcal{F}_i = \hat{F}_{\mathcal{S}_i}$
 I183: $\mathcal{S}_i \subset \mathcal{X}$
 I184: $\lim_{i \rightarrow \infty} \mu(\mathcal{F}_i) = \lim_{i \rightarrow \infty} \mu(\hat{F}_{\mathcal{S}_i}) = \mu(F(\mathcal{X}))$
 I185: $F(\cdot)$
 I186: $x, y \in \mathbb{R}$
 I187: $\theta \in \mathbb{R}$
 I188: $(u_v, u_\omega) \in \mathcal{U} \subset \mathbb{R}^2$
 I189: $\mathcal{X} \in \mathbb{R}^3$
 I190: T
 I191: ρ
 I192: u_v

$$\text{I193: } \theta_i = \frac{u_v t_i}{\rho}, \forall i \in \{1, 2, 3\}$$

$$\text{I194: } t_1$$

$$\text{I195: } t_2$$

$$\text{I196: } t_3$$

$$\text{I197: } t_1 + t_2 + t_3 \leq T$$

$$\text{I198: } u_v \in [v_{min}, v_{max}]$$

$$\text{I199: } 1$$

$$\text{I200: } (x, y)$$